# Racial Bias in Algorithmic Risk Assessment: An Analysis of the COMPAS Recidivism Scoring System

**Hia Pal | MSC Data Science | Madras School of Economics**

## Introduction

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) algorithm is a risk assessment tool used by judges and parole officers across the United States to predict the likelihood that a defendant will reoffend. Defendants are assigned a score from 1 to 10, with higher scores indicating greater predicted risk. These scores directly influence bail decisions, sentencing, and parole outcomes.

This project investigates whether the COMPAS algorithm exhibits racial bias by analyzing a dataset of 5,278 defendants from Broward County, Florida, originally compiled by ProPublica in 2016. The central question is: do Black defendants receive systematically higher risk scores than white defendants, even when their criminal histories are similar?

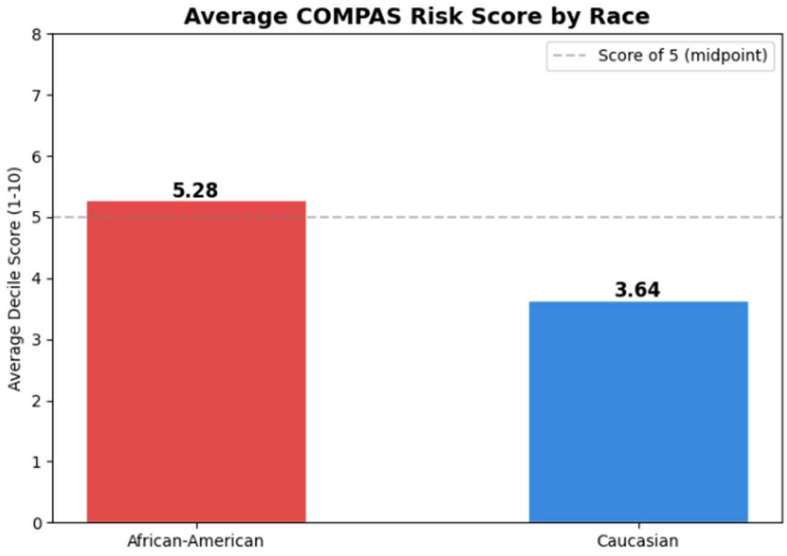
## Data & Methodology

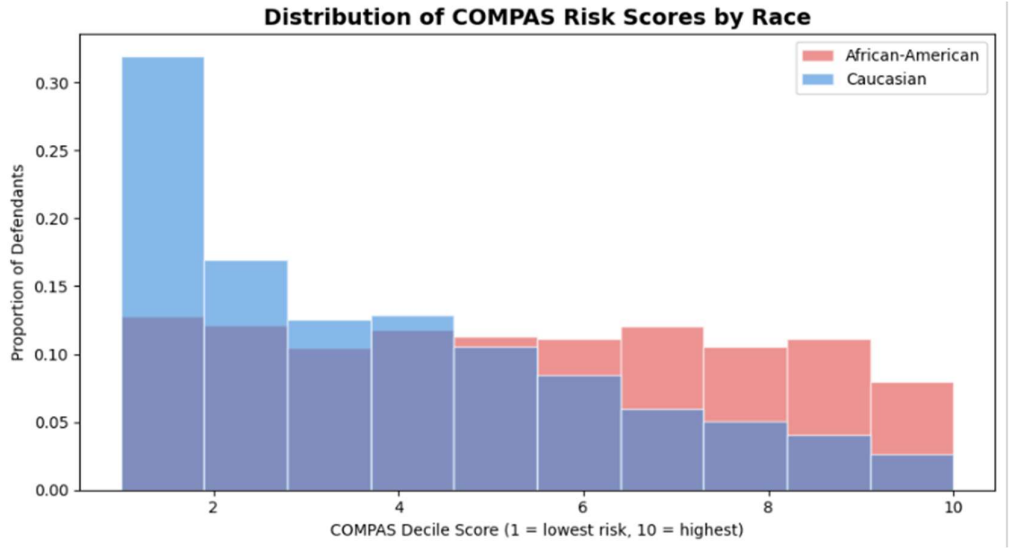
The dataset used in this analysis was sourced from ProPublica's COMPAS investigation, containing criminal records and COMPAS scores for defendants in Broward County, Florida between 2013 and 2014. The original dataset contained 7,214 records. After cleaning — removing cases where screening occurred more than 30 days after arrest, removing unknown recidivism flags, excluding traffic offenses, and focusing on African-American and Caucasian defendants — 5,278 records remained.

The analysis was conducted in Python using pandas for data manipulation and matplotlib and seaborn for visualization. No statistical model was built — this is an exploratory data analysis examining patterns in the raw data.

## Findings

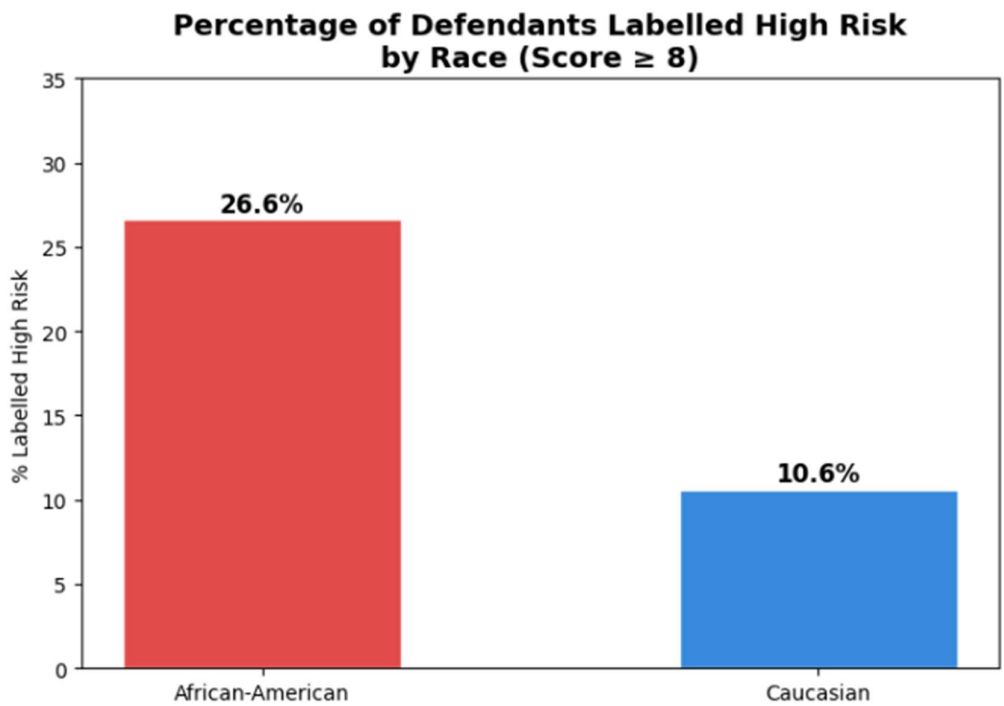
**Finding 1** — Higher average scores for Black defendants





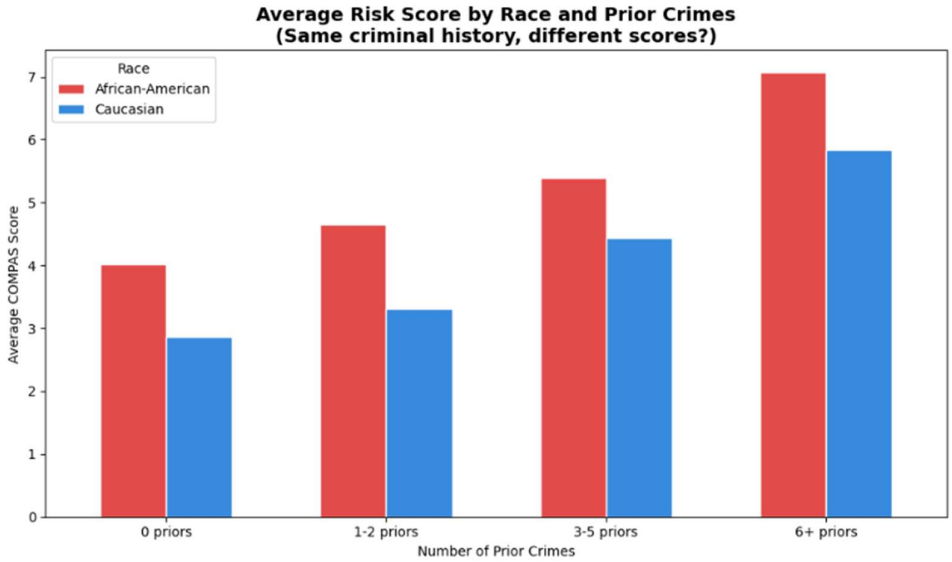

Black defendants received an average COMPAS risk score of 5.28 compared to 3.64 for white defendants — a gap of 1.64 points on a 10-point scale, representing a 45% higher average score. The distribution of scores (Chart 2) further shows that white defendants' scores cluster heavily at the low end (1–3), while Black defendants show a much more even spread across all risk levels including the high range of 8–10.

**Finding 2** — Black defendants flagged high risk at 2.5× the rate



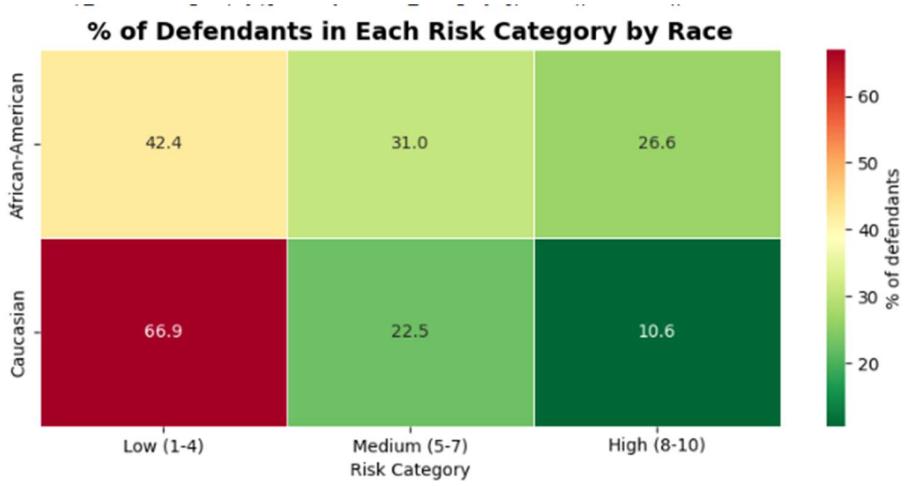

Using a threshold of 8 or above to define "high risk," 26.6% of Black defendants were labelled high risk compared to just 10.6% of white defendants. This means a Black defendant is 2.5 times more likely to be flagged as dangerous by the algorithm.

**Finding 3** — Disparity persists after controlling for criminal history

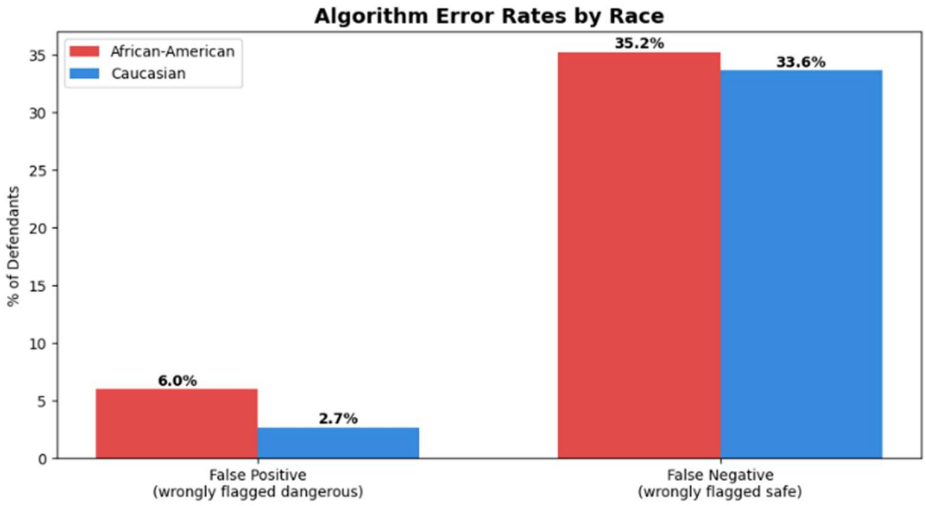



Crucially, when comparing defendants with the same number of prior crimes, Black defendants consistently receive higher scores than white defendants in every category — from 0 priors to 6 or more. This suggests that race itself, not just criminal history, is influencing the algorithm's output.

**Finding 4** — Unequal error rates Charts




The algorithm makes different kinds of errors depending on race. Black defendants face a false positive rate of 6.0% — meaning they were labelled high risk but did not go on to reoffend — compared to 2.7% for white defendants. This represents a 2.2× disparity in wrongful high-risk labelling.




## Limitations

This analysis has several important limitations. First, the dataset covers only one county in Florida and may not generalise to other jurisdictions. Second, the analysis focuses only on African-American and Caucasian defendants — other racial groups were excluded due to insufficient sample sizes. Third, while the data shows a clear correlation between race and risk scores, it does not prove causation or identify the exact mechanism by which bias enters the algorithm. Finally, several potentially relevant variables — such as quality of legal representation, socioeconomic status, and neighbourhood crime rates — are not present in the dataset and could partially explain the observed disparities.

# Conclusion

This analysis finds consistent evidence of racial bias in the COMPAS risk assessment algorithm. Black defendants receive significantly higher risk scores than white defendants with similar criminal histories, are flagged as high risk at 2.5 times the rate, and are wrongly labelled dangerous at more than twice the rate. Given that COMPAS scores directly influence real judicial decisions — including bail, sentencing, and parole — these disparities have serious consequences for Black defendants.

These findings suggest that algorithmic tools used in the criminal justice system require rigorous independent auditing for racial bias before deployment, and that human judges should exercise critical scrutiny when using algorithm-generated risk scores.